Deep Learning Practical Viva Questions

1. Study of Deep Learning Packages: TensorFlow, Keras, Theano, PyTorch
Conceptual Viva Questions:
1. What is TensorFlow?
2. What is Keras and how is it related to TensorFlow?
3. What is Theano?
4. What is PyTorch?
5. Difference between TensorFlow and PyTorch?
6. Why is Keras preferred by beginners?
7. What is a computation graph?
8. What are tensors?

2. Implementing Feedforward Neural Networks with Keras and TensorFlow
Conceptual Viva Questions:
1. What is a feedforward neural network (FNN)?
2. What is the MNIST dataset?
3. What is CIFAR-10?
4. What is the role of activation functions?
5. What is ReLU and Softmax?
6. What is SGD (Stochastic Gradient Descent)?
7. What are epochs and batch size?
8. Why do we normalize the dataset?
9. What is the loss function used in classification?
10. How do we check model accuracy?

3. Image Classification Model
Conceptual Viva Questions:
1. What is image classification?
2. What are the four stages in building an image classification model?
3. Why do we preprocess images?
4. What is the role of the model architecture?
5. What is overfitting? How can we prevent it?
6. Why do we split data into training and testing sets?
7. What is the use of accuracy and loss graphs?
8. Difference between training accuracy and validation accuracy?
9. What optimizer can be used other than SGD?
10. What happens if we don’t normalize image data?

4. Autoencoder for Anomaly Detection
Conceptual Viva Questions:
1. What is an Autoencoder?
2. What is anomaly detection?
3. How is Autoencoder used for anomaly detection?
4. What are the two main parts of an Autoencoder?
5. What does the Encoder do?
6. What does the Decoder do?
7. What is reconstruction error?
8. What loss function is typically used?
9. What optimizer is commonly used?
10. What is a latent representation?
11. What is ROC AUC in this context?

Bonus General Viva Questions:
1. Difference between supervised and unsupervised learning?

2. What is a neural network?
3. Why is GPU used in deep learning?
4. What is overfitting vs underfitting?
5. Difference between batch gradient descent and stochastic gradient descent?
6. What is a confusion matrix?
7. What is regularization and why is it used?
8. What is dropout in deep learning?
9. What is the role of activation functions like ReLU or Sigmoid?
10. What is the difference between CNN and Autoencoder?